

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/FINAL/D_DEB_AWKe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3303, C2 = 6636, C3 = 2929)

text_priors <- '
Distrib (Fm,          LogUniform,          0.0001,      0.3);
Distrib (pAm,         TruncNormal_cv,       100,         0.2,         40,       2000);
Distrib (r_pAm_pM,    TruncNormal_cv,       0.8,          0.1,         0.1,       2);
Distrib (v,           TruncNormal_cv,       0.18505,      0.1,         0.001,     2);
Distrib (kapJ,        Uniform,              0,            1);
Distrib (kapA,        Uniform,              0,            1);
Distrib (Ehp,         TruncNormal_cv,       100,         0.2,         20,       300);
Distrib (E_coc,       TruncNormal_cv,       30,          0.2,         1,       300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,     0.3);
Distrib (K,           Uniform,              0.0001,      1);
Distrib (dv,          TruncNormal_cv,       2,           0.5,         0,       10);
Distrib (de,          LogUniform,          0.0000000001, 0.1);
Distrib (tau,         TruncNormal_cv,       7,           0.5,         0,       50);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter, RTOL, ATOL)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kapJ.1.	kapA.1.
Result_mode	0.0006748850	160.85200	0.75549700	0.21207500	0.91353300	0.4941910
2.5%	0.0008929580	146.80100	0.61684300	0.15246520	0.87817600	0.4059780
97.5%	0.0043655600	187.56900	0.89550545	0.22223500	0.94290135	0.5720179
Min	0.0004348210	103.81200	0.54347900	0.11936000	0.02145000	0.3360560
Max	0.0244829000	205.91700	0.96271000	0.25693700	0.95477800	0.9053520
Result_mean	0.0025003353	164.76221	0.75725690	0.18782746	0.91364891	0.4906830
Result_sd	0.0009265459	10.28997	0.06935873	0.01782005	0.01823622	0.0437558
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	K.1.	dv.1.	
Result_mode	97.52490	27.452500	0.008774100	0.006119150	4.4772200	
2.5%	70.53390	23.467800	0.005956460	0.007693211	3.3838100	
97.5%	134.68200	36.154665	0.011013470	0.037700025	5.3240485	
Min	45.87010	20.835300	0.000421096	0.005260340	0.7356790	
Max	158.35100	42.157000	0.053683600	0.886490000	6.4585900	
Result_mean	100.87515	29.337502	0.008443047	0.020599375	4.3604728	
Result_sd	16.93349	3.324776	0.001311161	0.010703584	0.5126249	
	de.1.	tau.1.	Sigma_W.1.			
Result_mode	1.849940e-09	12.937100	0.12344800			
2.5%	1.376990e-09	7.738074	0.10925010			
97.5%	3.150146e-06	21.386780	0.13960200			
Min	1.001630e-09	0.391192	0.10006400			
Max	8.130040e-06	29.148600	6.29344000			
Result_mean	4.548596e-07	14.793606	0.12386119			
Result_sd	8.763850e-07	3.528219	0.05126382			

Number of observations, n = 277

Number of parameters, k = 13

Log-likelihood, LL = -28.04604

BIC = 129.2043

Potential scale reduction factors:

	Point est.	Upper C.I.
de.1.	1.03	1.05
dv.1.	1.05	1.15
E_coc.1.	1.02	1.05
Ehp.1.	1.02	1.07
Fm.1.	1.22	1.65
K.1.	1.15	1.43

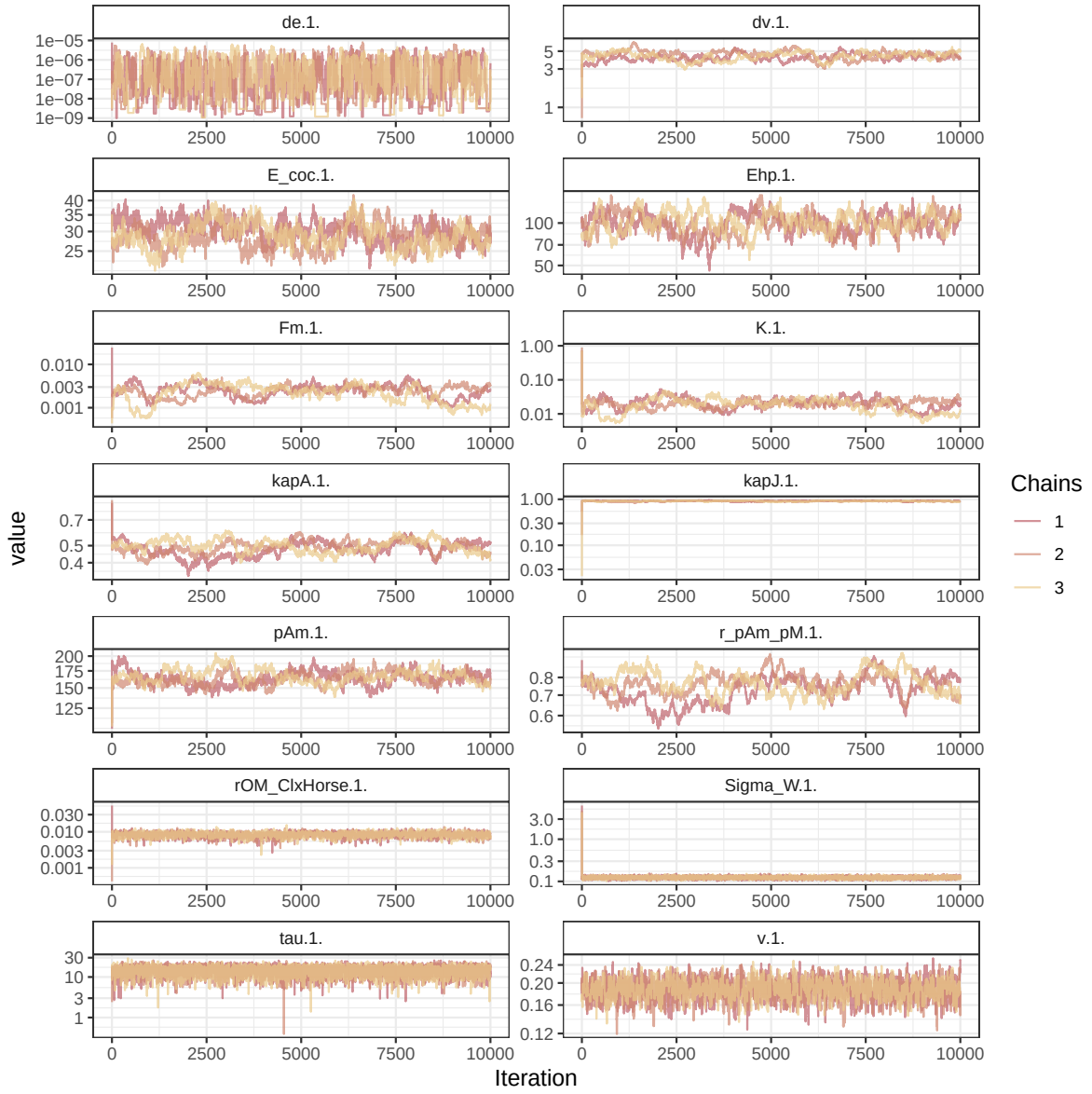


Figure 1: Parameters chains. Only the first iteration and the last `Nb_iter_kept` are represented.

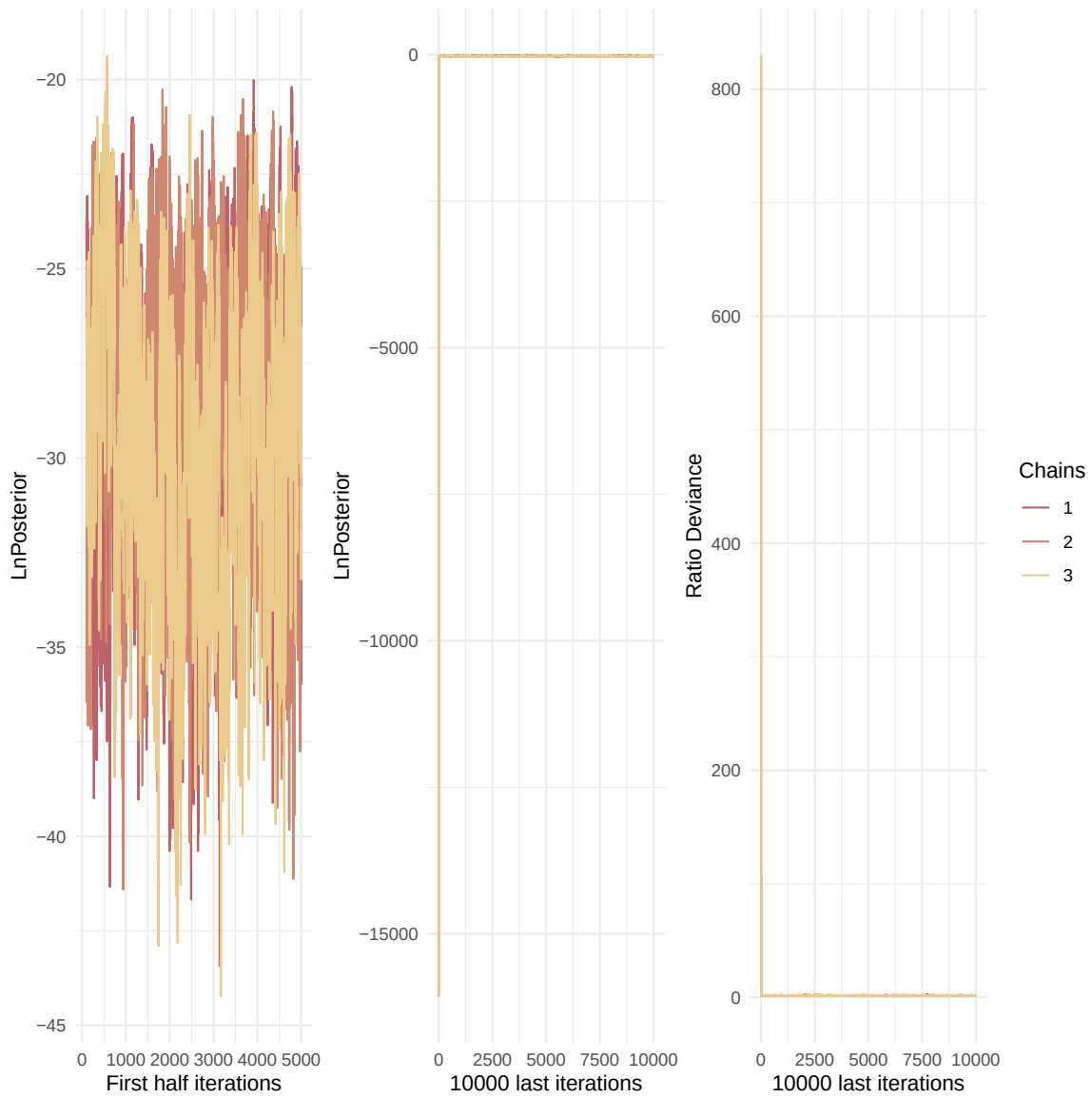


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

kapA.1.	1.03	1.09
kapJ.1.	1.06	1.20
pAm.1.	1.01	1.04
r_pAm_pM.1.	1.03	1.06
rOM_ClxHorse.1.	1.01	1.02
Sigma_W.1.	1.00	1.00
tau.1.	1.00	1.00
v.1.	1.01	1.02

Multivariate psrf

1.16

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

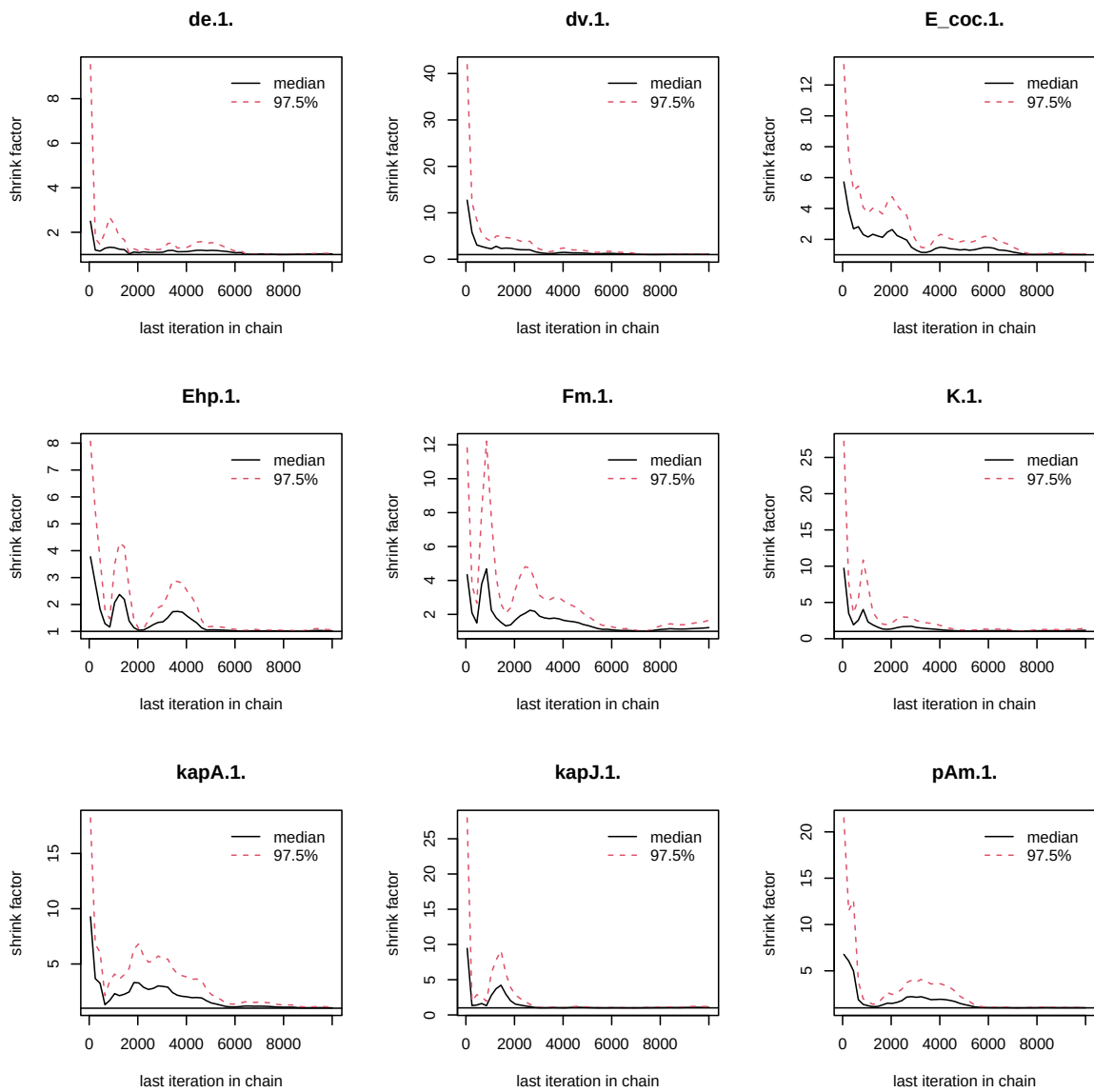


Figure 3: Chains convergence.

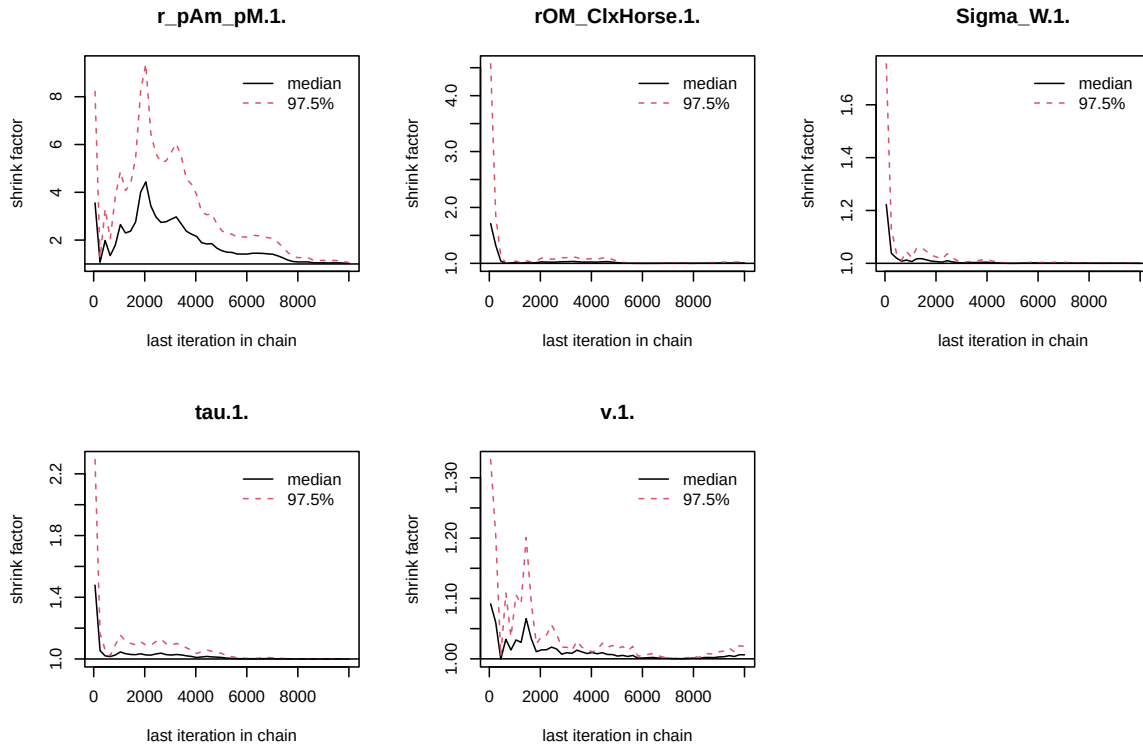


Figure 4: Chains convergence.

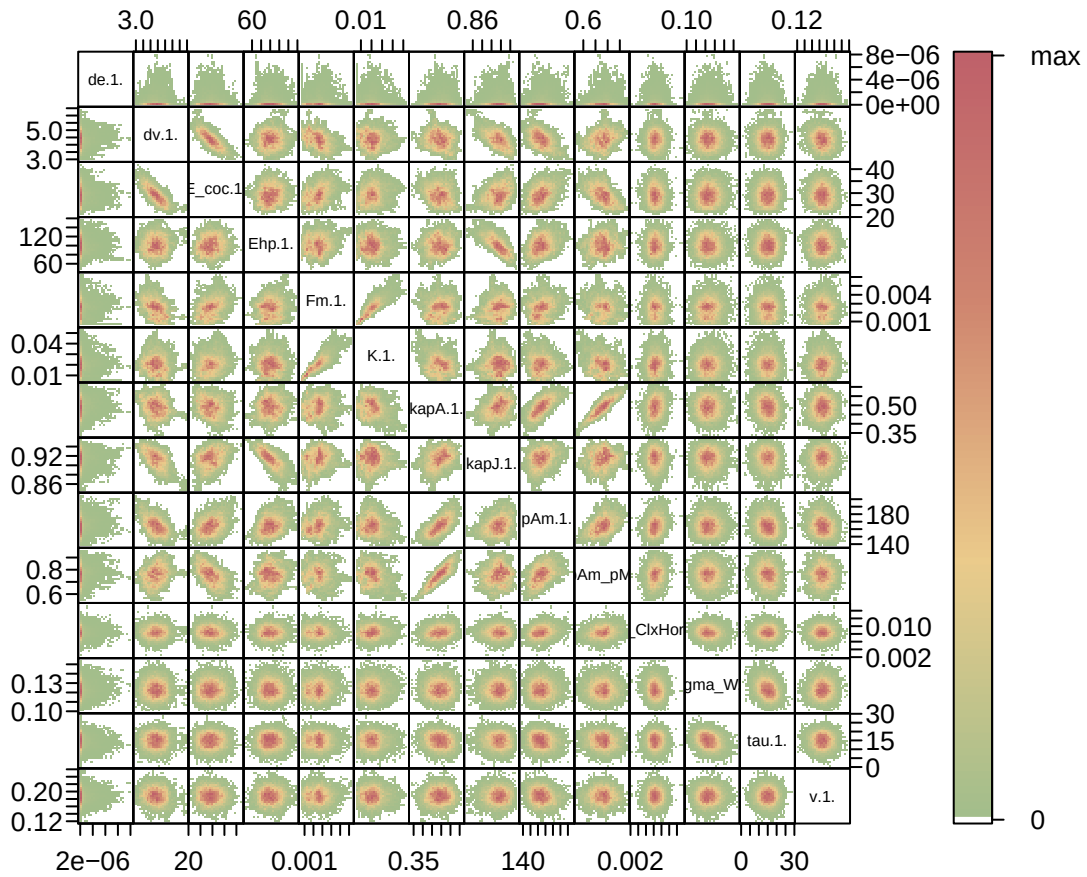


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

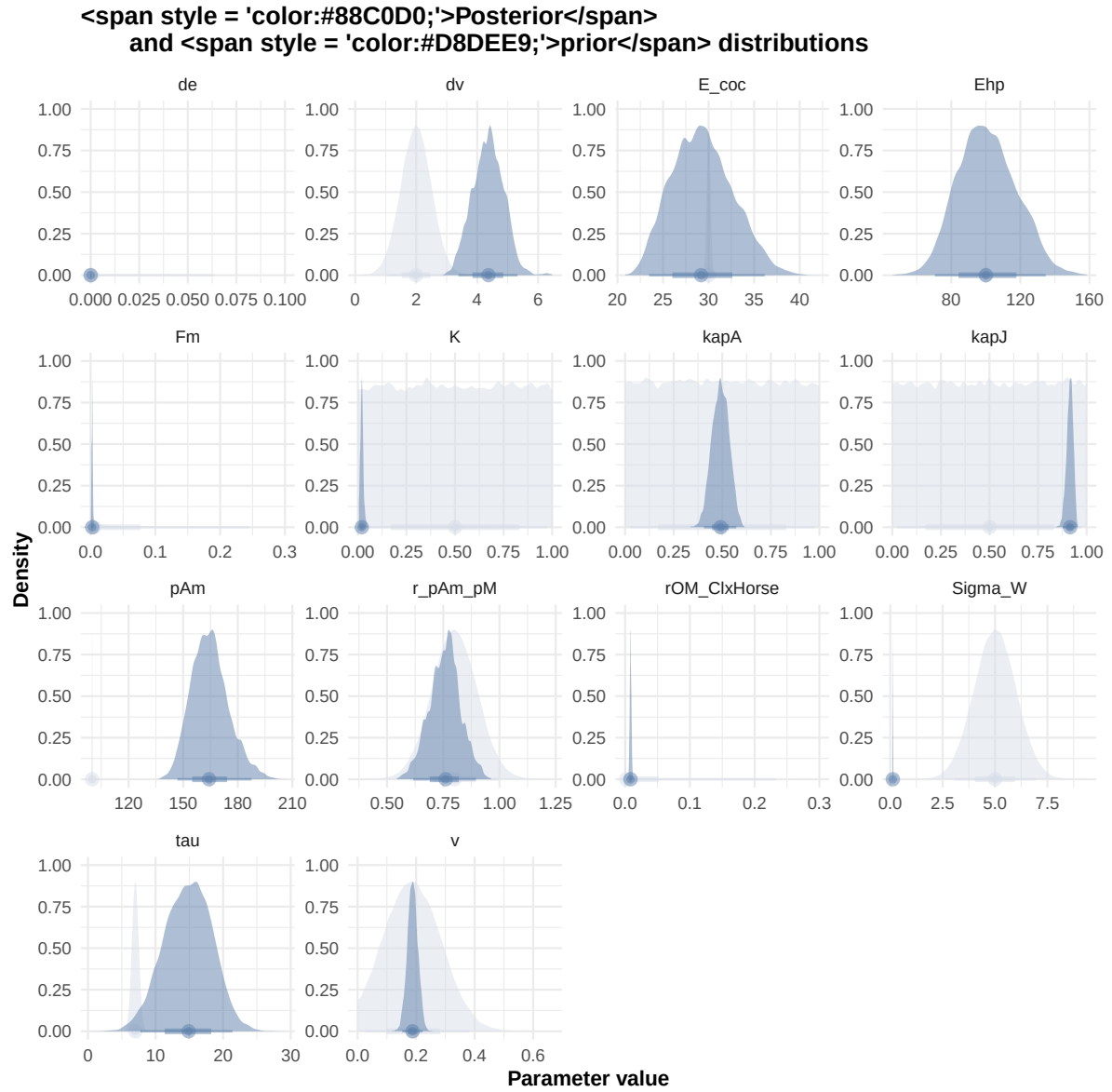


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  filter(Time %in% select_times) |>
  filter(!((Time > 70) & (ID_experiment == "Gollot2026_dens_D7"))) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
 i Row 1 of 'x' matches multiple rows in 'y'.
 i Row 1 of 'y' matches multiple rows in 'x'.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

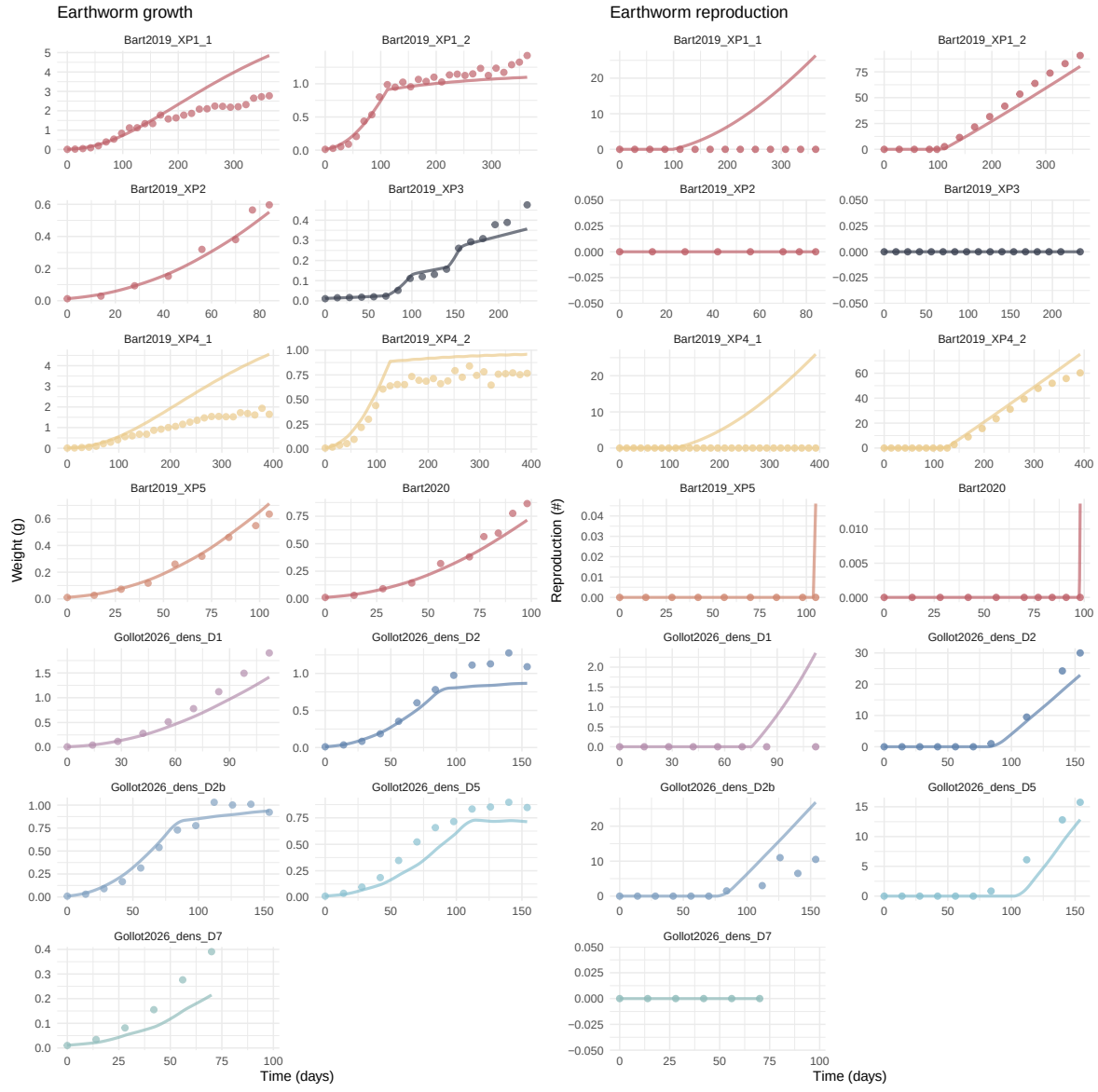
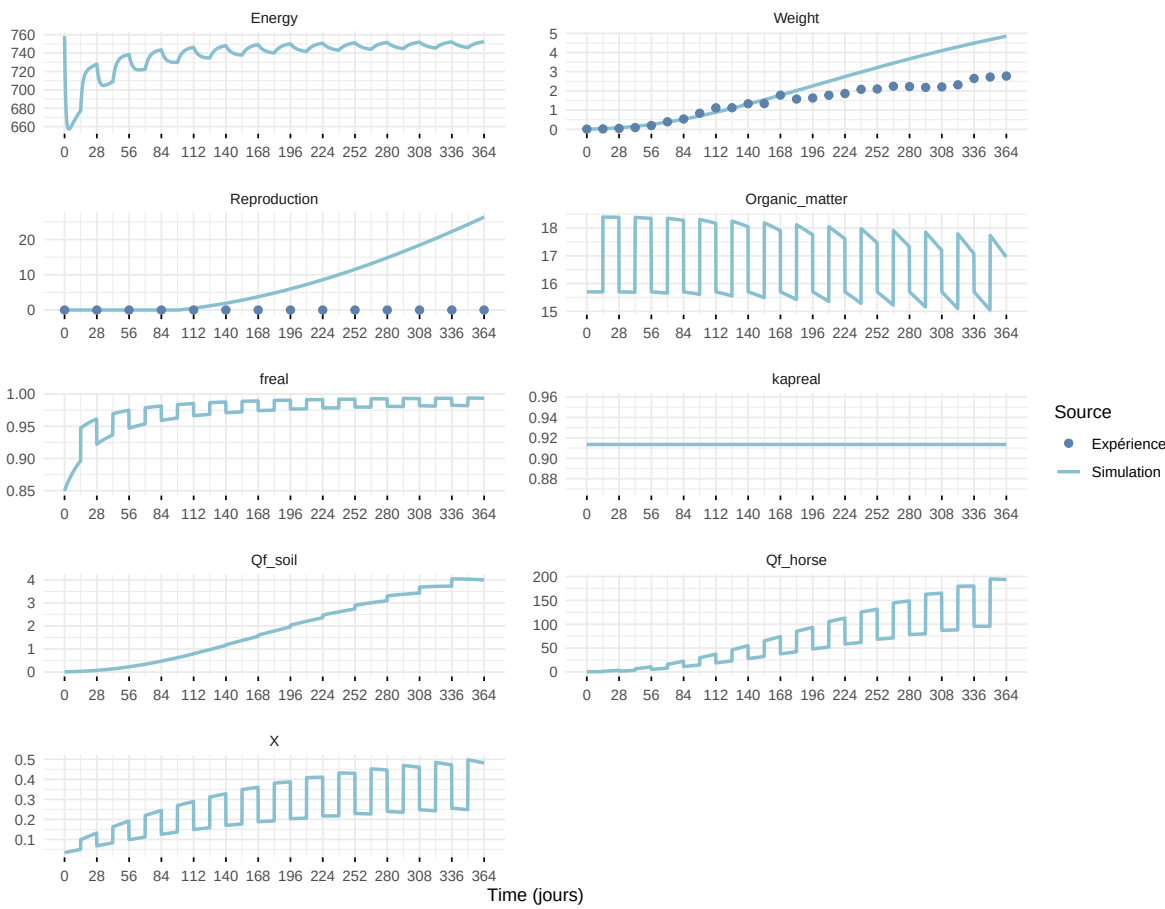
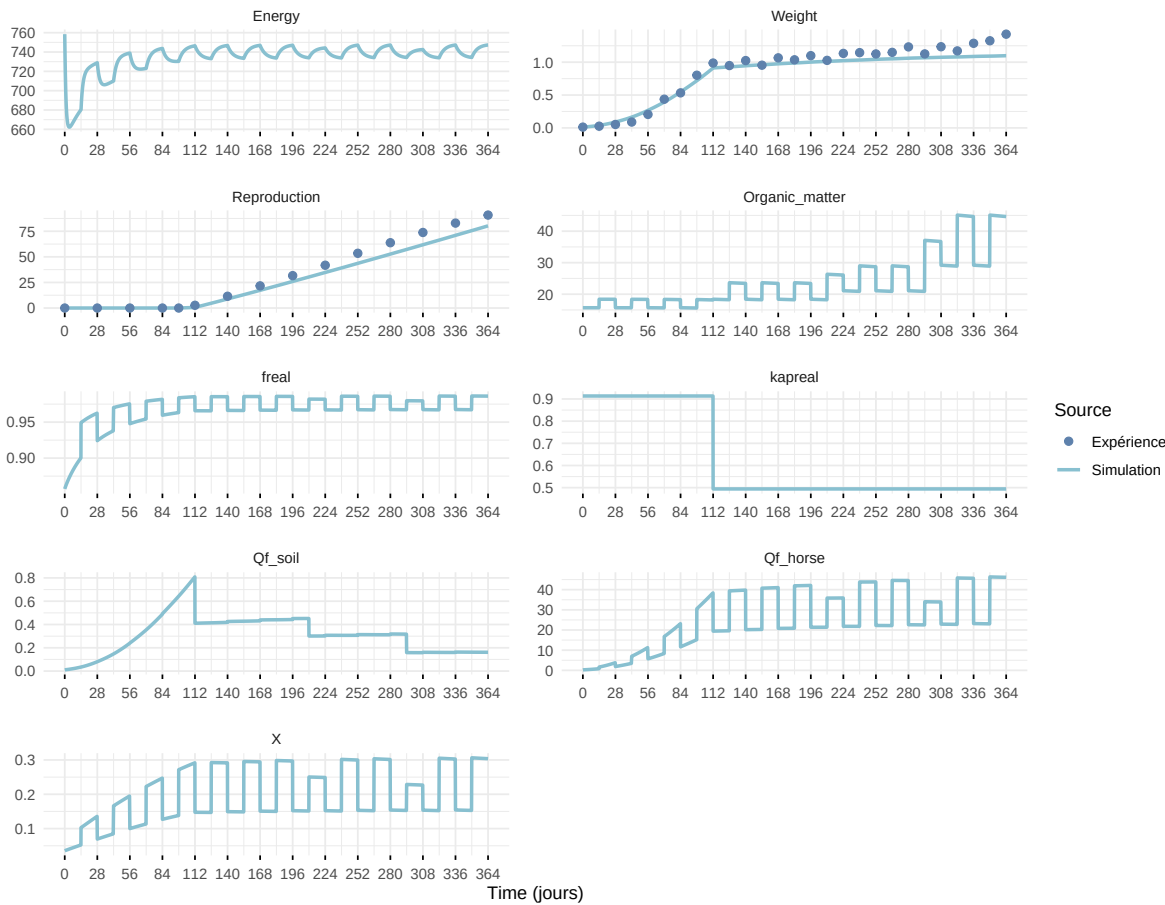


Figure 7: Simulations Weight and Reproduction.

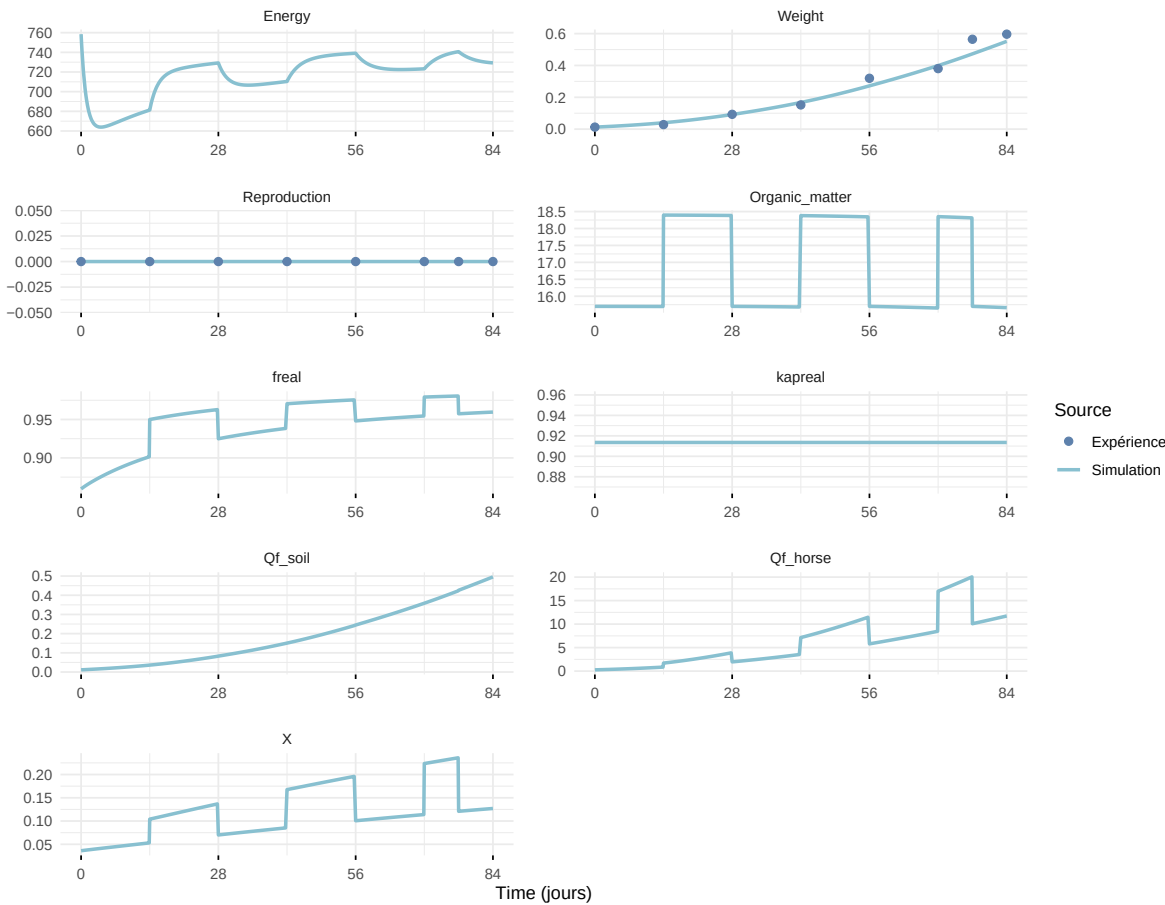
Expérience : Bart2019_XP1_1



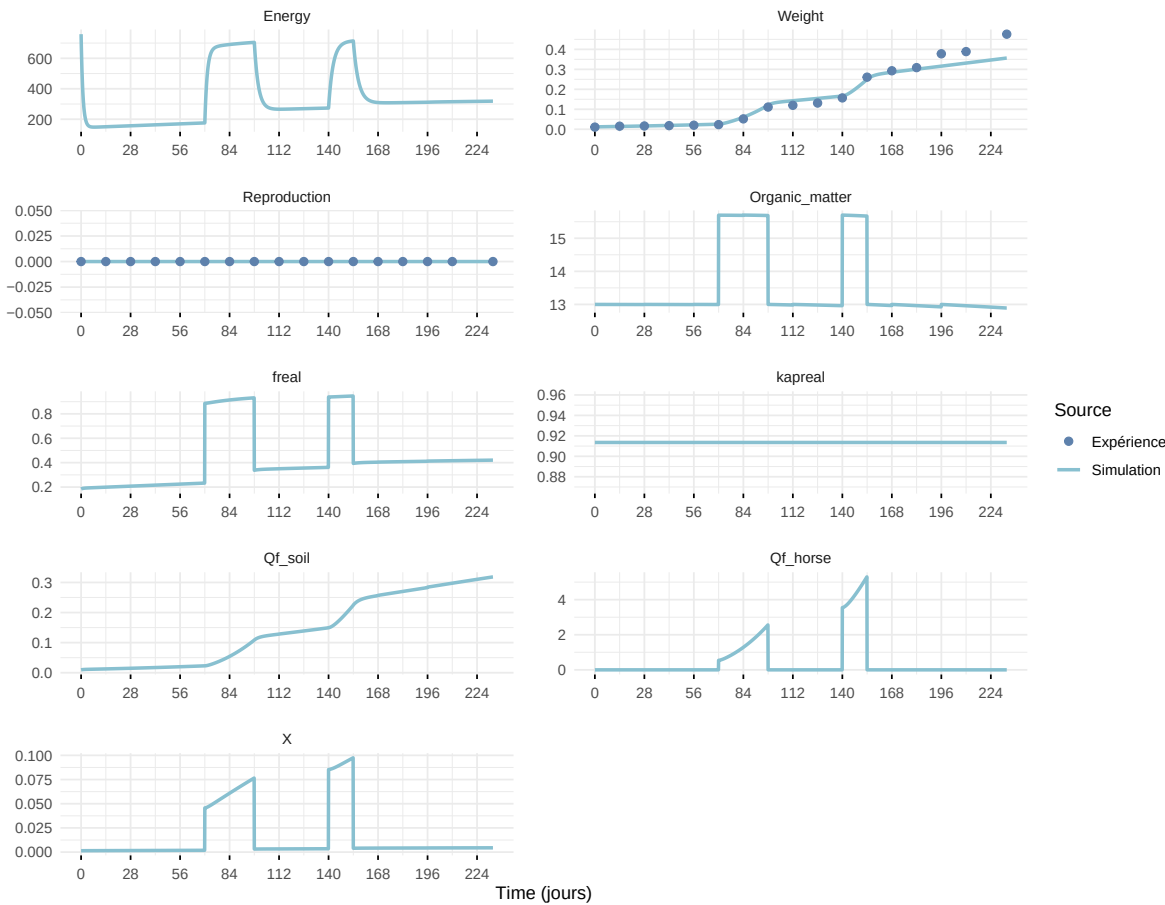
Expérience : Bart2019_XP1_2



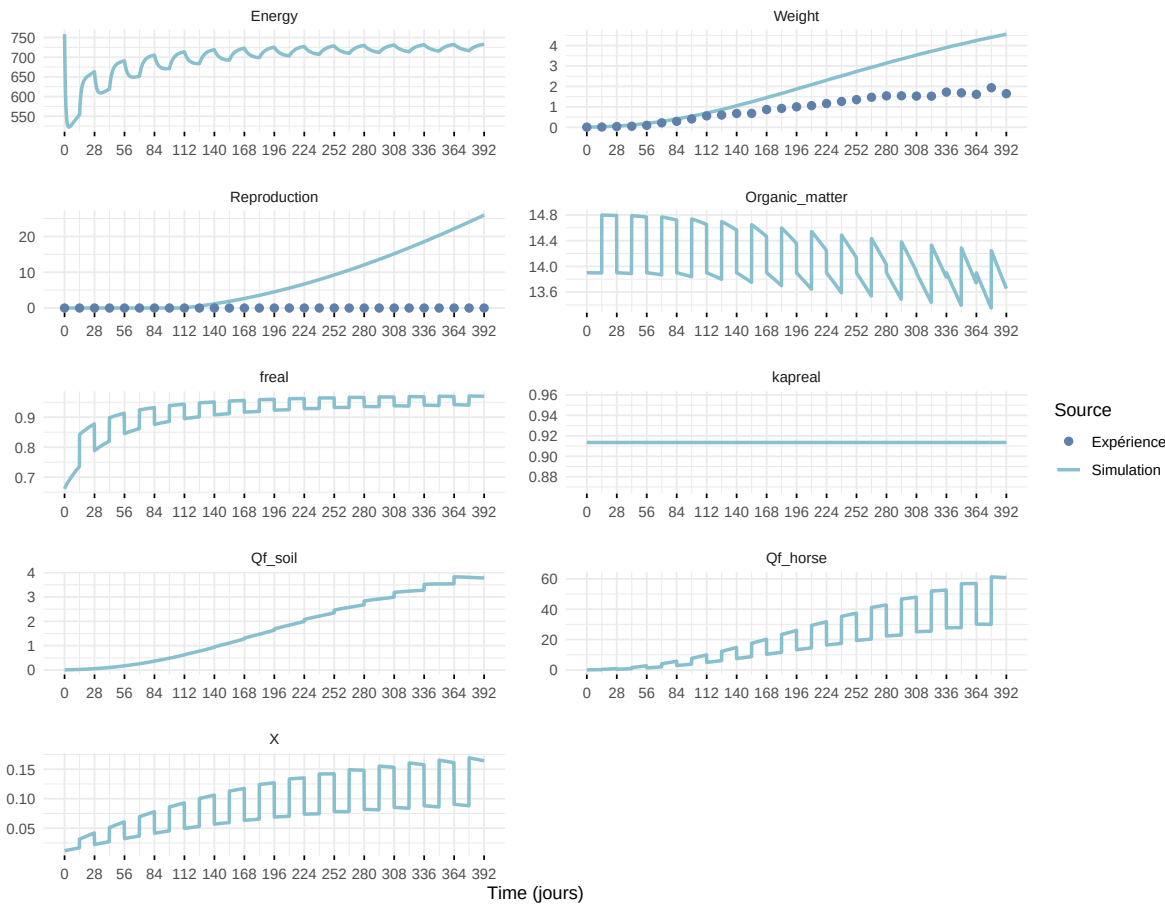
Expérience : Bart2019_XP2



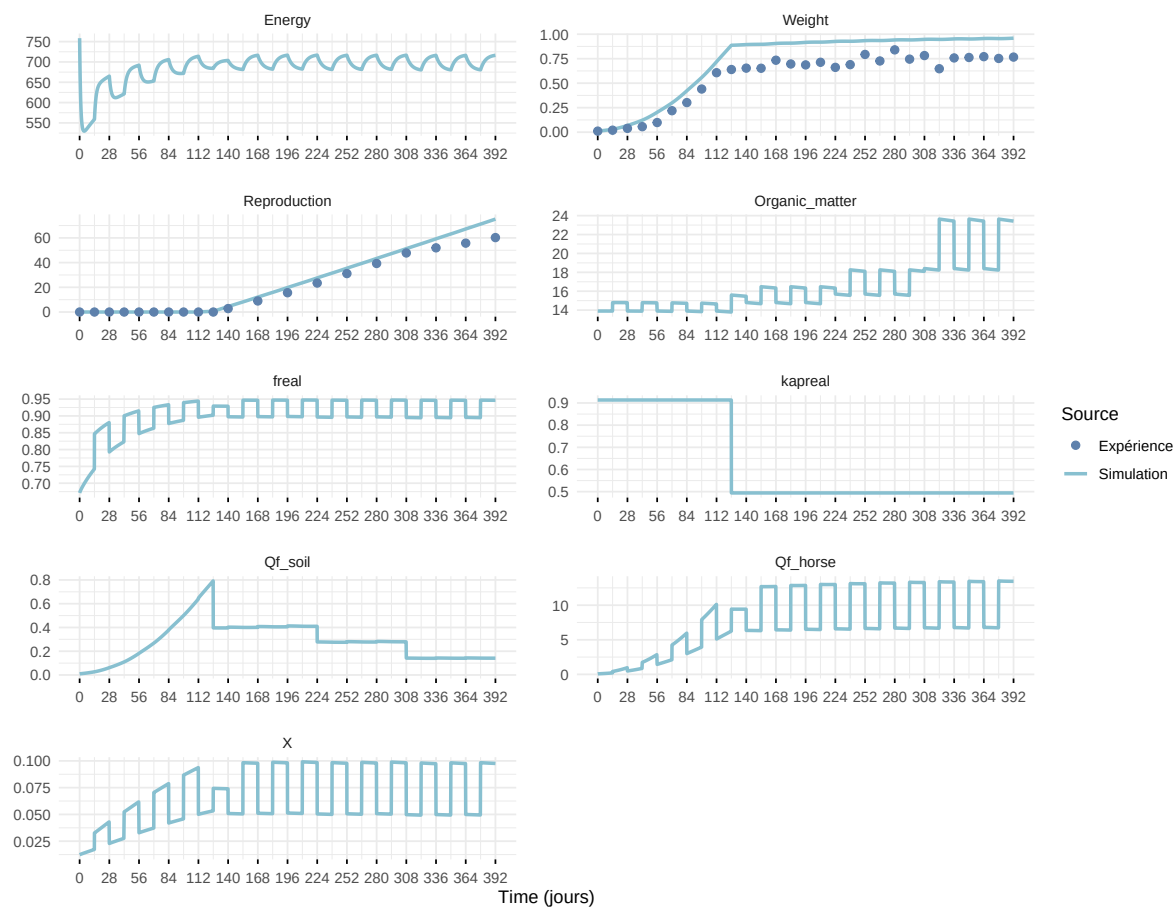
Expérience : Bart2019_XP3



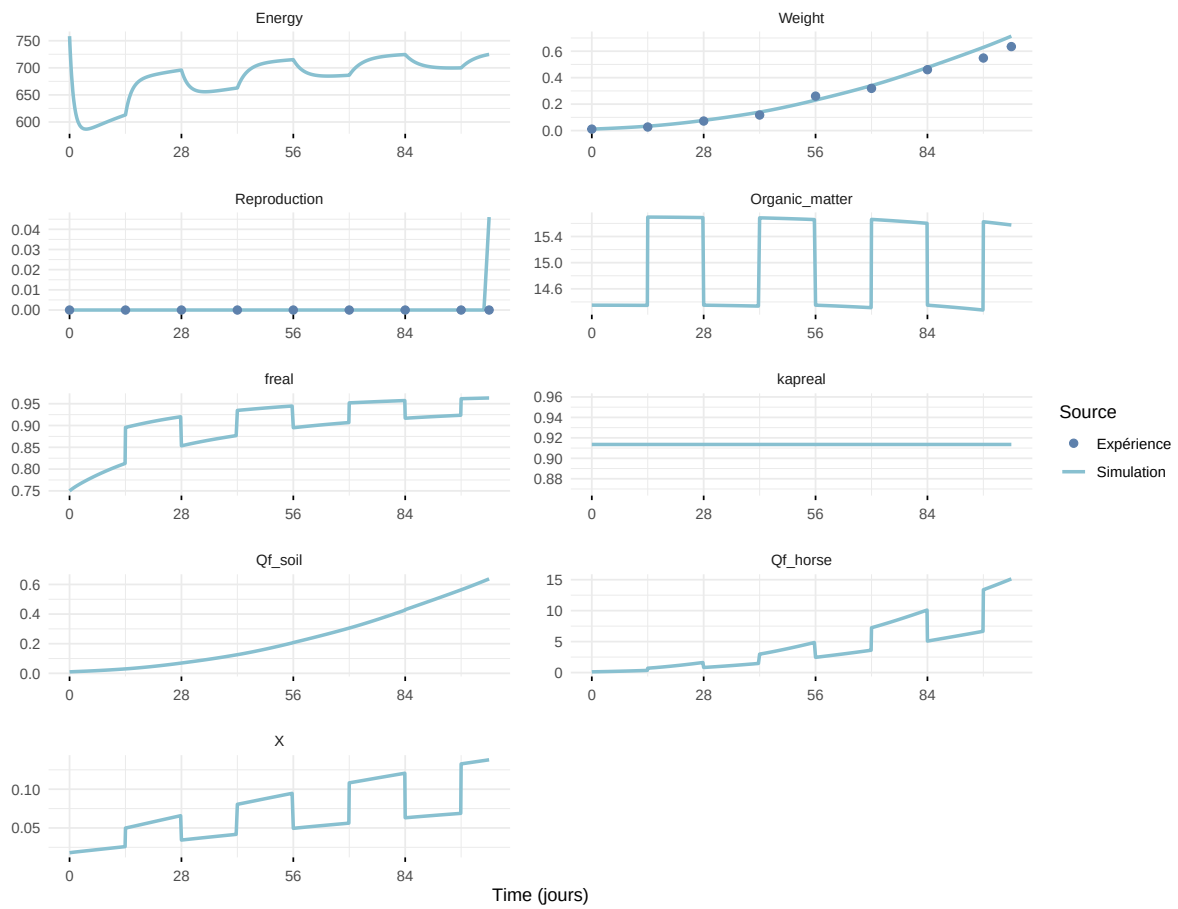
Expérience : Bart2019_XP4_1



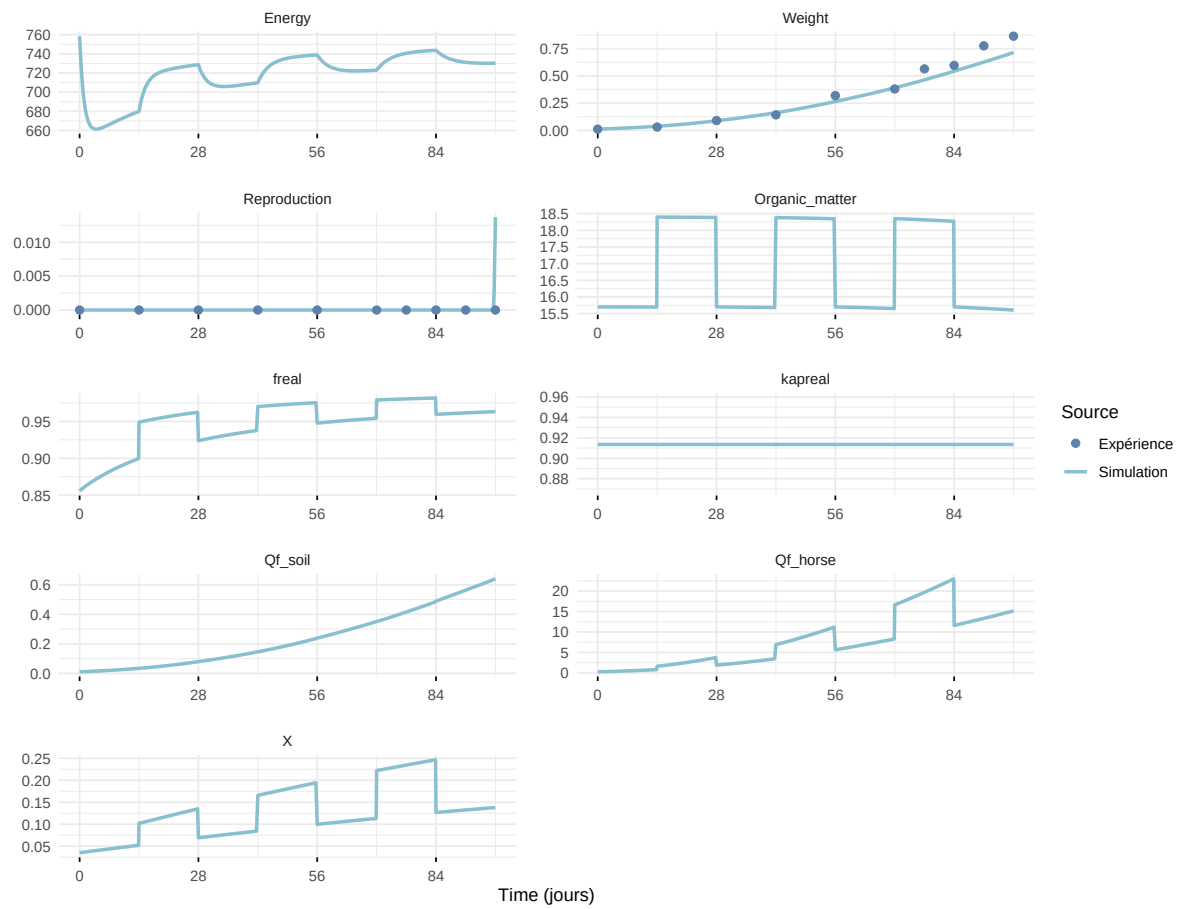
Expérience : Bart2019_XP4_2



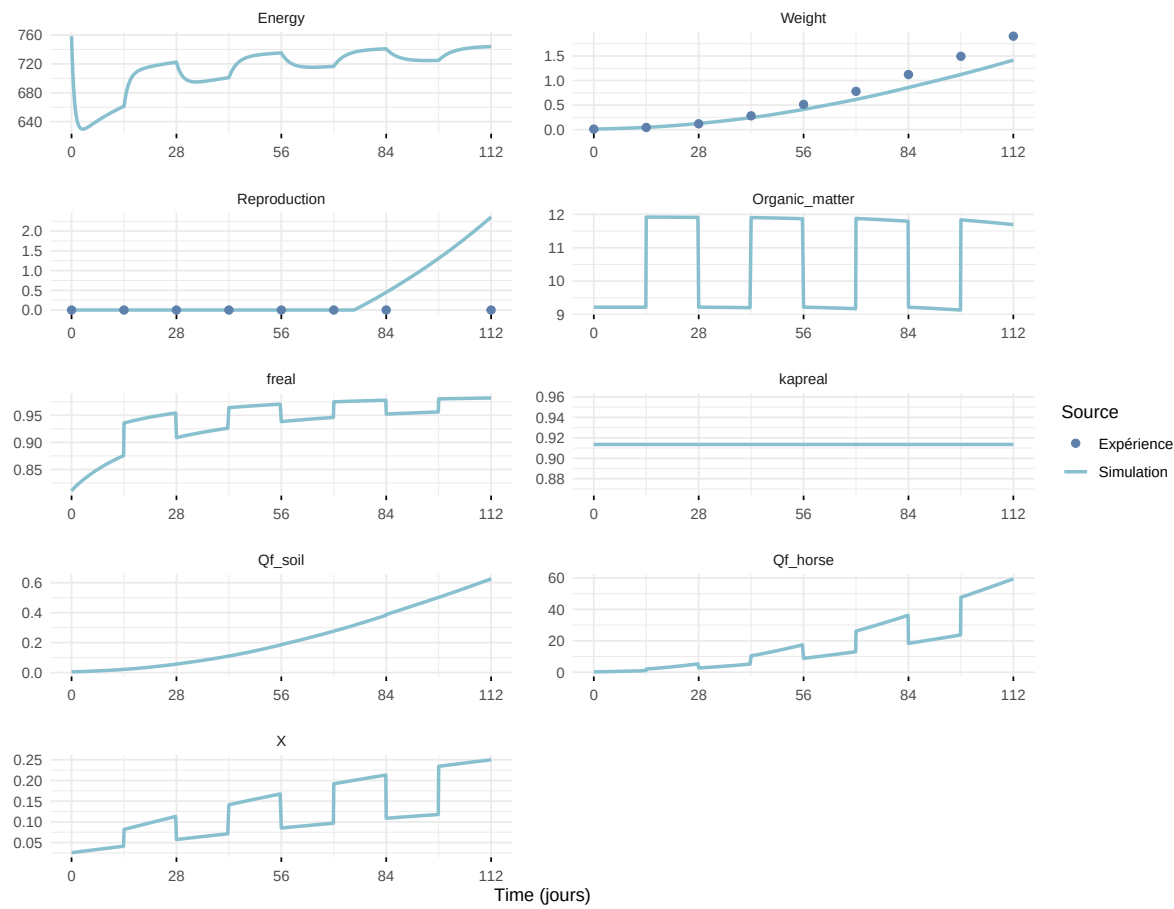
Expérience : Bart2019_XP5



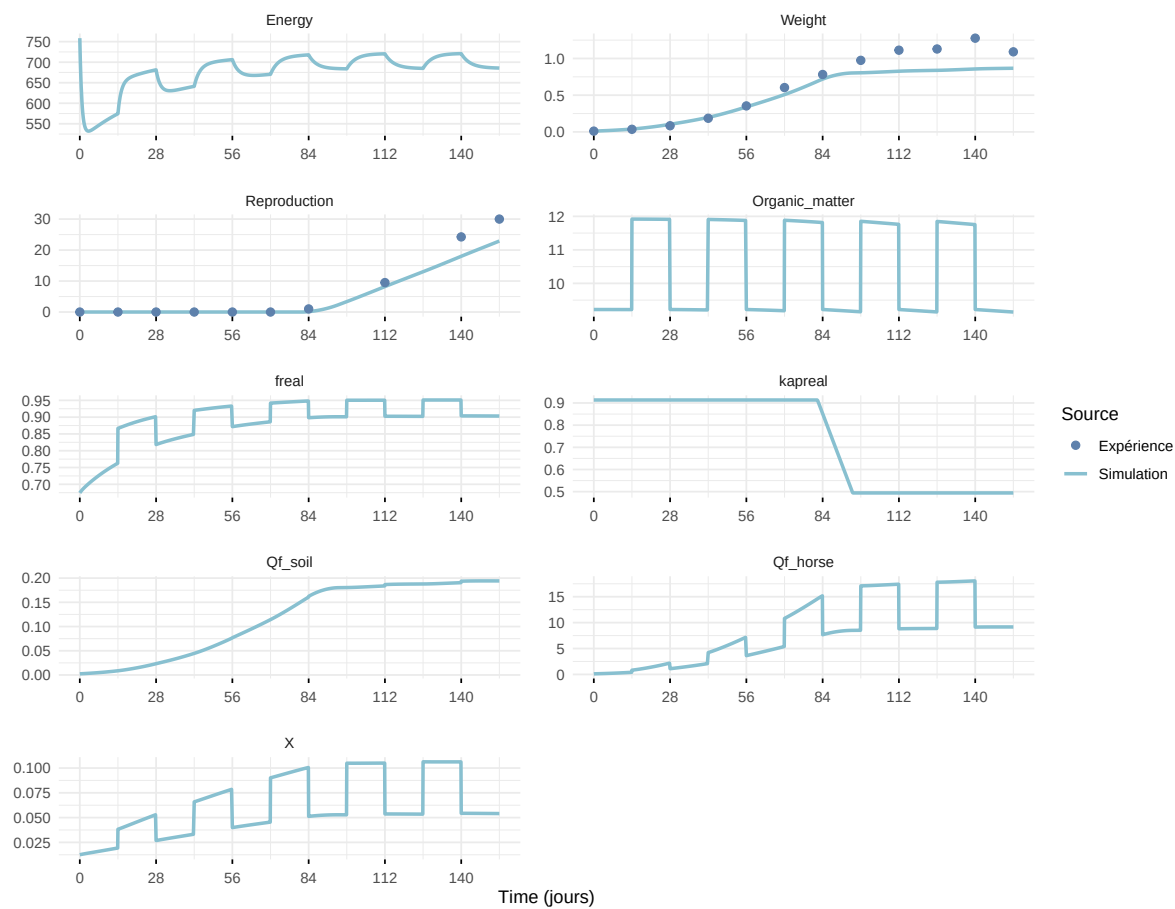
Expérience : Bart2020



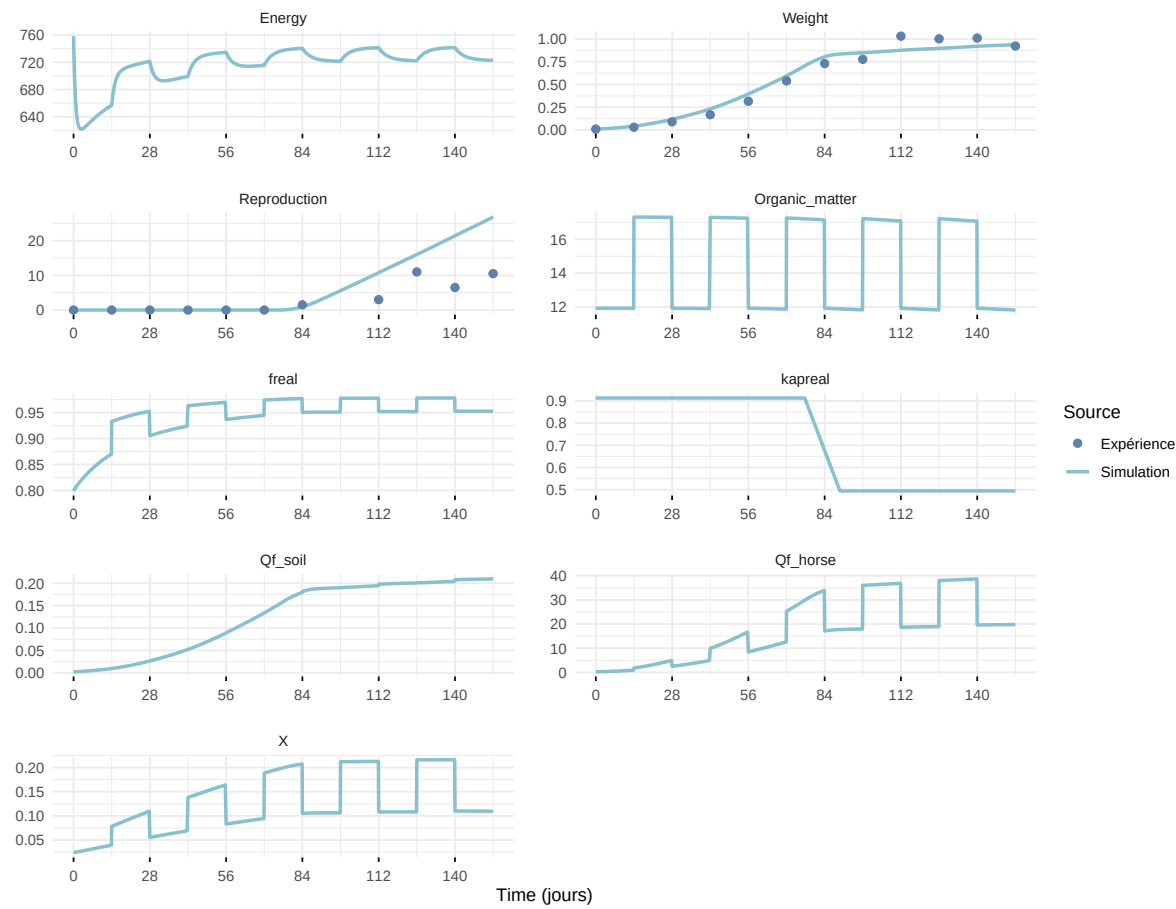
Expérience : Gollot2026_dens_D1



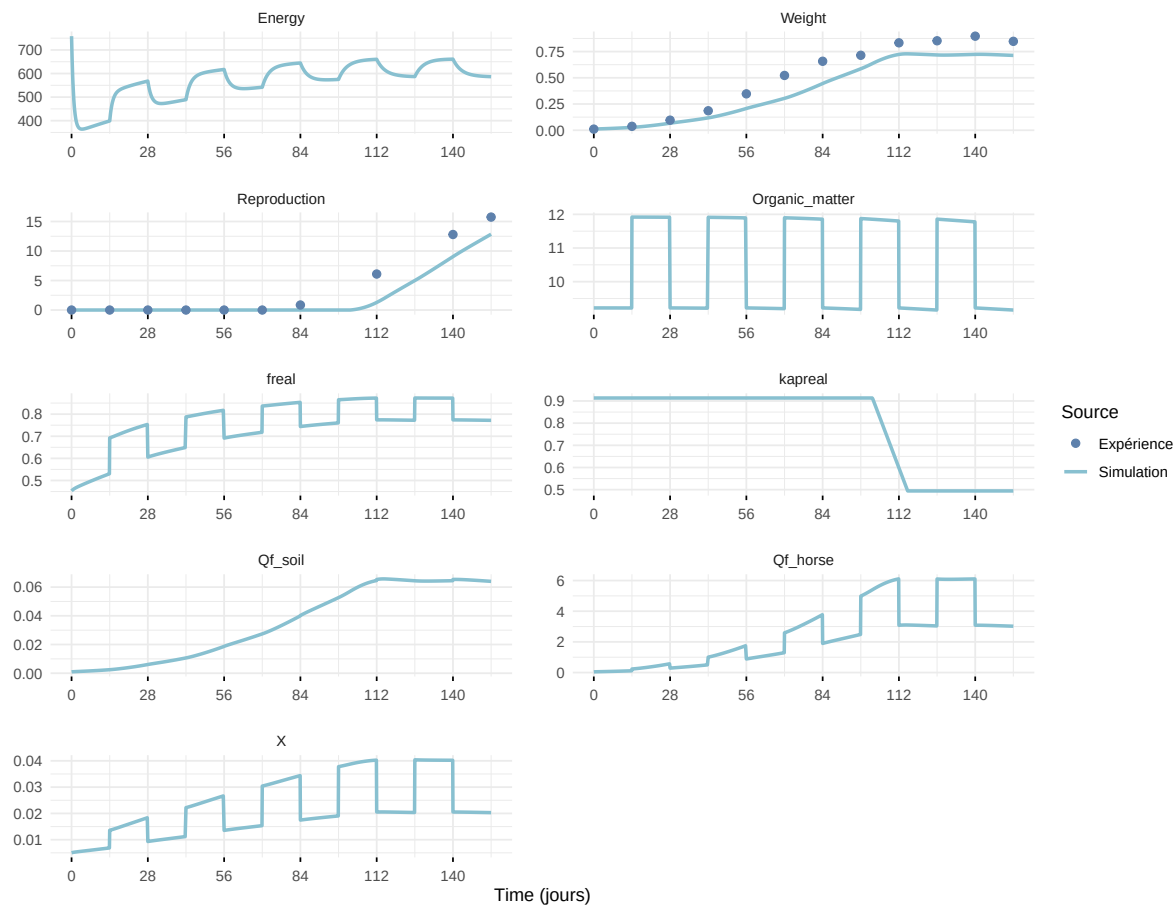
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

